

Optimization 2

ISE 5406

Lecture 11c: Accelerated Methods and Convergence Rate Comparisons

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Last Lectures

- Lecture 11: Steepest descent, Wolfe conditions, Newton's method
- Lecture 11b: Convergence of GD on L -smooth, μ -strongly convex functions
 - Descent lemma $\implies O(1/T)$ for convex
 - PL inequality $\implies (1 - \mu/L)^k$ linear rate for strongly convex

Today: Beyond Gradient Descent

Can we do **better** than $(1 - 1/\kappa)^k$ where $\kappa = L/\mu$?

- Newton's method: **quadratic** local convergence
- Conjugate gradient: rate $\left(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}\right)^k$ on quadratics
- Heavy ball method: rate $\approx 1 - 2/\sqrt{\kappa}$ on quadratics

Newton's Method: Recap

Given $f : \mathbb{R}^n \rightarrow \mathbb{R}$ twice continuously differentiable:

Newton Update

$$\mathbf{x}_{k+1} = \mathbf{x}_k - (\nabla^2 f(\mathbf{x}_k))^{-1} \nabla f(\mathbf{x}_k)$$

Idea: At each step, minimize the **quadratic Taylor approximation**

$$q_k(\mathbf{x}) = f(\mathbf{x}_k) + \nabla f(\mathbf{x}_k)^T (\mathbf{x} - \mathbf{x}_k) + \frac{1}{2} (\mathbf{x} - \mathbf{x}_k)^T \nabla^2 f(\mathbf{x}_k) (\mathbf{x} - \mathbf{x}_k)$$

Key advantage: Affine-invariant (independent of coordinate scaling)

Key disadvantage: Requires computing/inverting the Hessian ($O(n^3)$ per step)

Local Quadratic Convergence

Theorem (Local Quadratic Convergence of Newton's Method)

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be twice continuously differentiable.

Assume:

1. $\nabla^2 f(\mathbf{x}) \succeq \mu \mathbf{I}$ for all $\mathbf{x}$ (strongly convex with parameter $\mu > 0$)
2. $\|\nabla^2 f(\mathbf{x}) - \nabla^2 f(\mathbf{y})\| \leq L_H \|\mathbf{x} - \mathbf{y}\|$ for all $\mathbf{x}, \mathbf{y}$ (Lipschitz Hessian)

Then for the sequence $\{\mathbf{x}_k\}$ generated by Newton's method:

$$\|\mathbf{x}_{k+1} - \mathbf{x}^*\| \leq \frac{L_H}{2\mu} \|\mathbf{x}_k - \mathbf{x}^*\|^2$$

Proof of Quadratic Convergence

Step 1: Since $\nabla f(\mathbf{x}^*) = \mathbf{0}$:

$$\begin{aligned}\mathbf{x}_{k+1} - \mathbf{x}^* &= \mathbf{x}_k - \mathbf{x}^* - (\nabla^2 f(\mathbf{x}_k))^{-1} \nabla f(\mathbf{x}_k) \\ &= (\nabla^2 f(\mathbf{x}_k))^{-1} [\nabla^2 f(\mathbf{x}_k)(\mathbf{x}_k - \mathbf{x}^*) - (\nabla f(\mathbf{x}_k) - \nabla f(\mathbf{x}^*))]\end{aligned}$$

Step 2: Use the **integral form** of the mean value theorem:

$$\nabla f(\mathbf{x}_k) - \nabla f(\mathbf{x}^*) = \int_0^1 \nabla^2 f(\mathbf{x}^* + t(\mathbf{x}_k - \mathbf{x}^*))(\mathbf{x}_k - \mathbf{x}^*) dt$$

Step 3: Substitute:

$$\mathbf{x}_{k+1} - \mathbf{x}^* = (\nabla^2 f(\mathbf{x}_k))^{-1} \int_0^1 [\nabla^2 f(\mathbf{x}_k) - \nabla^2 f(\mathbf{x}^* + t(\mathbf{x}_k - \mathbf{x}^*))](\mathbf{x}_k - \mathbf{x}^*) dt$$

Step 4: Take norms using $\|(\nabla^2 f(\mathbf{x}_k))^{-1}\| \leq 1/\mu$ and Lipschitz Hessian:

$$\|\mathbf{x}_{k+1} - \mathbf{x}^*\| \leq \frac{1}{\mu} \int_0^1 L_H (1-t) \|\mathbf{x}_k - \mathbf{x}^*\| dt = \frac{L_H}{2\mu} \|\mathbf{x}_k - \mathbf{x}^*\|^2$$

Newton with Backtracking: Global Convergence

Problem: Pure Newton can diverge if started far from $\mathbf{x}^*$.

Fix: **Damped Newton** — use backtracking line search on the Newton direction.

Damped Newton Method

1. Compute Newton direction: $\mathbf{d}_k = -(\nabla^2 f(\mathbf{x}_k))^{-1} \nabla f(\mathbf{x}_k)$
2. Choose step size t_k via backtracking (Armijo condition):

$$f(\mathbf{x}_k + t_k \mathbf{d}_k) \leq f(\mathbf{x}_k) + \alpha t_k \nabla f(\mathbf{x}_k)^T \mathbf{d}_k$$

3. Update: $\mathbf{x}_{k+1} = \mathbf{x}_k + t_k \mathbf{d}_k$

Two-Phase Convergence

- **Phase 1** (far from $\mathbf{x}^*$): backtracking reduces step, ensures

Newton: Practical Implications

Phase	Convergence	Per-step cost
Global (backtracking)	Linear	$O(n^3)$
Local (full step)	Quadratic	$O(n^3)$

When to use Newton:

- n is moderate (Hessian inversion is $O(n^3)$)
- High accuracy is needed (quadratic convergence $\implies$ few iterations suffice)
- Hessian is available / cheap to compute

Iteration count comparison (to reach ϵ accuracy, $\kappa = L/\mu$):

- Gradient descent: $O(\kappa \log(1/\epsilon))$ iterations
- Newton (near optimum): $O(\log \log(1/\epsilon))$ iterations!

CG Method: Recap

For the quadratic $f(\mathbf{x}) = \frac{1}{2}\mathbf{x}^T \mathbf{Q} \mathbf{x} - \mathbf{b}^T \mathbf{x}$ with $\mathbf{Q} \succ 0$:

CG Update

$$\mathbf{x}_{k+1} = \mathbf{x}_k + \alpha_k \mathbf{p}_k$$

$$\mathbf{r}_{k+1} = \mathbf{r}_k + \alpha_k \mathbf{Q} \mathbf{p}_k$$

$$\mathbf{p}_{k+1} = -\mathbf{r}_{k+1} + \gamma_k \mathbf{p}_k$$

$$\text{where } \alpha_k = \frac{\mathbf{r}_k^T \mathbf{r}_k}{\mathbf{p}_k^T \mathbf{Q} \mathbf{p}_k}, \quad \gamma_k = \frac{\mathbf{r}_{k+1}^T \mathbf{r}_{k+1}}{\mathbf{r}_k^T \mathbf{r}_k}$$

Key properties (from Lecture 11):

- Directions $\mathbf{p}_0, \mathbf{p}_1, \dots$ are $\mathbf{Q}$ -conjugate: $\mathbf{p}_i^T \mathbf{Q} \mathbf{p}_j = 0$ for $i \neq j$
- **Finite termination:** Exact solution in $\leq n$ steps
- $\mathbf{x}_k$ minimizes f over $\mathbf{x}_0 + \text{span}\{\mathbf{r}_0, \mathbf{Q} \mathbf{r}_0, \dots, \mathbf{Q}^{k-1} \mathbf{r}_0\}$ (Krylov)

CG Convergence Rate

Theorem (Convergence Rate of CG)

Let $\mathbf{Q}$ have eigenvalues $0 < \mu = \lambda_1 \leq \dots \leq \lambda_n = L$ with condition number $\kappa = L/\mu$. Then:

$$\|\mathbf{x}_k - \mathbf{x}^*\|_{\mathbf{Q}} \leq 2 \left(\frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1} \right)^k \|\mathbf{x}_0 - \mathbf{x}^*\|_{\mathbf{Q}}$$

where $\|\mathbf{u}\|_{\mathbf{Q}} = \sqrt{\mathbf{u}^T \mathbf{Q} \mathbf{u}}$ is the energy norm.

Key insight: The rate depends on $\sqrt{\kappa}$, not κ itself!

Example: $\kappa = 10,000$

- Steepest descent: needs $O(\kappa \log(1/\epsilon)) = O(10,000 \cdot \log(1/\epsilon))$ iterations

Proof Sketch: CG Rate via Chebyshev Polynomials

Step 1: Krylov subspace property: $\mathbf{x}_k$ minimizes f over $\mathbf{x}_0 + \mathcal{K}_k(\mathbf{Q}, \mathbf{r}_0)$.

Step 2: Polynomial formulation:

$$\|\mathbf{x}_k - \mathbf{x}^*\|_{\mathbf{Q}} = \min_{\substack{p \in \mathcal{P}_k \\ p(0)=1}} \|p(\mathbf{Q})(\mathbf{x}_0 - \mathbf{x}^*)\|_{\mathbf{Q}}$$

where $\mathcal{P}_k$ = polynomials of degree $\leq k$ with $p(0) = 1$.

Step 3: Eigenvalue bound: Since $\mathbf{Q} = U\Lambda U^T$:

$$\|p(\mathbf{Q})\mathbf{u}\|_{\mathbf{Q}}^2 = \sum_{i=1}^n \lambda_i (p(\lambda_i))^2 c_i^2 \leq \max_{\lambda \in [\mu, L]} p(\lambda)^2 \cdot \|\mathbf{u}\|_{\mathbf{Q}}^2$$

Step 4: Chebyshev optimality: The polynomial that minimizes $\max_{\lambda \in [\mu, L]} |p(\lambda)|$ subject to $p(0) = 1$ is the **scaled Chebyshev polynomial**:

Momentum: The Key Idea

Gradient descent uses only the current gradient:

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \alpha \nabla f(\mathbf{x}_k)$$

Can we use **memory of past iterates** to accelerate?

ODE Analogy: Gradient Flow with Friction

$$\ddot{\mathbf{x}}(t) + \gamma \dot{\mathbf{x}}(t) + \nabla f(\mathbf{x}(t)) = 0 \quad (\text{damped oscillator})$$

Discretizing with time step Δt gives two-term recurrences:

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \alpha \nabla f(\mathbf{x}_k) + \beta(\mathbf{x}_k - \mathbf{x}_{k-1})$$

The **momentum term** $\beta(\mathbf{x}_k - \mathbf{x}_{k-1})$ carries forward previous velocity.

Heavy Ball Method (Polyak, 1964)

Heavy Ball Update

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \alpha \nabla f(\mathbf{x}_k) + \beta (\mathbf{x}_k - \mathbf{x}_{k-1})$$

with $\mathbf{x}_{-1} := \mathbf{x}_0$.

Optimal parameters for $f(\mathbf{x}) = \frac{1}{2}\mathbf{x}^T \mathbf{Q}\mathbf{x} - \mathbf{b}^T \mathbf{x}$ with $\mu \leq \lambda_i \leq L$:

$$\alpha = \frac{4}{(\sqrt{L} + \sqrt{\mu})^2}, \quad \beta = \left(\frac{\sqrt{L} - \sqrt{\mu}}{\sqrt{L} + \sqrt{\mu}} \right)^2 = \left(\frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1} \right)^2$$

Theorem (Heavy Ball Convergence on Quadratics)

With optimal α, β :

$$\|\mathbf{x}_k - \mathbf{x}^*\| \leq C \left(\frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1} \right)^k \|\mathbf{x}_0 - \mathbf{x}^*\|$$

Proof: Heavy Ball via Spectral Analysis

Step 1: Error recurrence. Let $\mathbf{e}_k = \mathbf{x}_k - \mathbf{x}^*$. Since $\nabla f(\mathbf{x}_k) = \mathbf{Q}\mathbf{e}_k$:

$$\mathbf{e}_{k+1} = (1 + \beta)\mathbf{e}_k - \alpha\mathbf{Q}\mathbf{e}_k - \beta\mathbf{e}_{k-1}$$

Write as a **companion system**:

$$\begin{pmatrix} \mathbf{e}_{k+1} \\ \mathbf{e}_k \end{pmatrix} = \underbrace{\begin{pmatrix} (1+\beta)I - \alpha\mathbf{Q} & -\beta I \\ I & 0 \end{pmatrix}}_T \begin{pmatrix} \mathbf{e}_k \\ \mathbf{e}_{k-1} \end{pmatrix}$$

Step 2: Block diagonalization. Via $\mathbf{Q} = U\Lambda U^T$, the matrix T decouples into n independent 2×2 blocks:

$$T_i = \begin{pmatrix} (1 + \beta)(1 - \alpha\lambda_i) & -\beta \\ 1 & 0 \end{pmatrix}, \quad i = 1, \dots, n$$

Proof (cont'd): Spectral Radius

Step 4: For optimal parameters and any $\lambda_i \in [\mu, L]$:

$$|\nu_{i,1}| = |\nu_{i,2}| = \sqrt{\beta} = \frac{\sqrt{L} - \sqrt{\mu}}{\sqrt{L} + \sqrt{\mu}} = \frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1}$$

Therefore the **spectral radius** of T is:

$$\rho(T) = \max_i \max_j |\nu_{i,j}| = \sqrt{\beta} = \frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1}$$

Step 5: By **Gelfand's formula**: $\rho(T) = \lim_{k \rightarrow \infty} \|T^k\|^{1/k}$

For any $\epsilon > 0$, there exists C_ϵ such that $\|T^k\| \leq C_\epsilon(\rho(T) + \epsilon)^k$.

Since $\mathbf{w}_k = T^k \mathbf{w}_0$ where $\mathbf{w}_k = (\mathbf{e}_k, \mathbf{e}_{k-1})^T$:

$$\|\mathbf{x}_k - \mathbf{x}^*\| \leq \|\mathbf{w}_k\| \leq C_\epsilon \left(\frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1} + \epsilon \right)^k \|\mathbf{w}_0\| \quad \square$$

Heavy Ball: Limitations

Warning: Heavy Ball Can Diverge!

The convergence guarantee for the heavy ball method is **only for quadratics**.

With fixed parameters α, β , the heavy ball method can **diverge** on non-quadratic strongly convex functions.

Practical issues:

- Requires knowledge of μ and L to set optimal parameters
- Not robust to misspecification of the condition number
- Oscillatory behavior near the optimum (complex eigenvalues)

Motivation for Nesterov: Can we get the $\sqrt{\kappa}$ improvement **globally** for all smooth strongly convex functions?

Nesterov's Method (Nesterov, 1983)

Nesterov's Accelerated Gradient (Strongly Convex)

$$\mathbf{y}_k = \mathbf{x}_k + \beta_k(\mathbf{x}_k - \mathbf{x}_{k-1}) \quad (\text{momentum / "look-ahead"})$$

$$\mathbf{x}_{k+1} = \mathbf{y}_k - \frac{1}{L} \nabla f(\mathbf{y}_k) \quad (\text{gradient step from } \mathbf{y}_k)$$

with $\beta_k = \frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}$ and $\mathbf{x}_{-1} := \mathbf{x}_0$.

Key difference from Heavy Ball:

- Gradient is evaluated at the **extrapolated point** $\mathbf{y}_k$, not at $\mathbf{x}_k$
- This "look-ahead" makes the method **globally convergent**

Parameters: $\alpha_k = 1/L$, $\beta_k = \frac{\sqrt{L}-\sqrt{\mu}}{\sqrt{L}+\sqrt{\mu}} = \frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}$

Nesterov: Convergence on Strongly Convex Functions

Theorem (Nesterov AGD — Strongly Convex Case)

Let f be L -smooth and μ -strongly convex with $\kappa = L/\mu$.

With parameters $\alpha_k = 1/L$ and $\beta_k = \frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}$:

$$f(\mathbf{x}_k) - f^* \leq \left(1 - \frac{1}{\sqrt{\kappa}}\right)^k \left[f(\mathbf{x}_0) - f^* + \frac{\mu}{2} \|\mathbf{x}_0 - \mathbf{x}^*\|^2 \right]$$

Iteration complexity to reach ϵ -accuracy:

$$k = O\left(\sqrt{\kappa} \log \frac{1}{\epsilon}\right)$$

Compare with GD: $k = O(\kappa \log \frac{1}{\epsilon})$

Proof via Lyapunov Function (Wright & Recht, 2022)

Step 1: Define the **Lyapunov function** with $\rho^2 = 1 - 1/\sqrt{\kappa}$:

$$V_k = f(\mathbf{x}_k) - f^* + \frac{L}{2} \|\tilde{\mathbf{x}}_k - \rho^2 \tilde{\mathbf{x}}_{k-1}\|^2$$

where $\tilde{\mathbf{x}}_k = \mathbf{x}_k - \mathbf{x}^*$.

Step 2: Goal: Show $V_{k+1} \leq \rho^2 V_k$.

Key inequalities used:

(i) **Descent lemma** (step $1/L$):

$$f(\mathbf{x}_{k+1}) \leq f(\mathbf{y}_k) - \frac{1}{2L} \|\nabla f(\mathbf{y}_k)\|^2$$

(ii) **Strong convexity:**

$$f(\mathbf{x}^*) \geq f(\mathbf{y}_k) + \nabla f(\mathbf{y}_k)^T (\mathbf{x}^* - \mathbf{y}_k) + \frac{\mu}{2} \|\mathbf{x}^* - \mathbf{y}_k\|^2$$

(iii) **Smoothness:**

$$f(\mathbf{y}_k) \leq f(\mathbf{x}_k) + \nabla f(\mathbf{y}_k)^T (\mathbf{y}_k - \mathbf{x}_k) + \frac{L}{2} \|\mathbf{y}_k - \mathbf{x}_k\|^2$$

Proof (cont'd): Lyapunov Contraction

Step 3: Expand V_{k+1} using inequality (i) and the Nesterov update $\mathbf{x}_{k+1} = \mathbf{y}_k - \frac{1}{L} \nabla f(\mathbf{y}_k)$:

$$V_{k+1} \leq f(\mathbf{y}_k) - \frac{1}{2L} \|\nabla f(\mathbf{y}_k)\|^2 - f^* + \frac{L}{2} \|\tilde{\mathbf{x}}_{k+1} - \rho^2 \tilde{\mathbf{x}}_k\|^2$$

Step 4: Apply inequalities (ii) and (iii) with the substitution $\mathbf{u}_k = \frac{1}{L} \nabla f(\mathbf{y}_k)$:

After algebraic manipulation (grouping terms involving $f(\mathbf{x}_k) - f^*$, $f(\mathbf{y}_k) - f^*$, and the quadratic terms), one obtains:

$$V_{k+1} \leq \rho^2 \left[f(\mathbf{x}_k) - f^* + \frac{L}{2} \|\tilde{\mathbf{x}}_k - \rho^2 \tilde{\mathbf{x}}_{k-1}\|^2 \right] + R_k = \rho^2 V_k + R_k$$

where the remainder R_k can be shown to be ≤ 0 for the specific choice of $\rho^2 = 1 - 1/\sqrt{\kappa}$.

Nesterov: Convex (Non-Strongly-Convex) Case

For convex f (i.e., $\mu = 0$), we need **variable** momentum parameters.

Nesterov AGD — Convex Version

$$\begin{aligned}\mathbf{y}_k &= \mathbf{x}_k + \beta_k(\mathbf{x}_k - \mathbf{x}_{k-1}) \\ \mathbf{x}_{k+1} &= \mathbf{y}_k - \frac{1}{L} \nabla f(\mathbf{y}_k)\end{aligned}$$

where $\beta_k = \frac{\rho_{k-1}^2(1-\rho_{k-1}^2)}{\rho_{k-1}^2 + \rho_k^2}$ and ρ_k satisfies
 $1 + \rho_k(\rho_{k-1}^2 - 1) - \rho_k^2 = 0$.

Theorem (Nesterov AGD — Convex Case)

$$f(\mathbf{x}_k) - f^* \leq \frac{2L\|\mathbf{x}_0 - \mathbf{x}^*\|^2}{(k+1)^2}$$

The Big Picture: Strongly Convex Functions

Method	Rate (per iteration)	Iterations for ϵ
Gradient Descent	$1 - \frac{2}{\kappa+1}$	$O(\kappa \log \frac{1}{\epsilon})$
Heavy Ball (quadratic)	$\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1} \approx 1 - \frac{2}{\sqrt{\kappa}}$	$O(\sqrt{\kappa} \log \frac{1}{\epsilon})$
CG (quadratic)	$2\left(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}\right)^k$	$O(\sqrt{\kappa} \log \frac{1}{\epsilon})$
Nesterov AGD	$1 - \frac{1}{\sqrt{\kappa}}$	$O(\sqrt{\kappa} \log \frac{1}{\epsilon})$
Newton (local)	Quadratic	$O(\log \log \frac{1}{\epsilon})$

Key takeaway: Momentum methods improve $\kappa \rightarrow \sqrt{\kappa}$ in the iteration count.

The Big Picture: Convex Functions

Method	Rate
Gradient Descent	$O(1/k)$
Nesterov AGD	$O(1/k^2)$

Lower bound (Nesterov, 2004): For any first-order method using only gradient evaluations, the worst-case rate on L -smooth convex functions is:

$$f(\mathbf{x}_k) - f^* \geq \frac{3L\|\mathbf{x}_0 - \mathbf{x}^*\|^2}{32(k+1)^2}$$

Nesterov's method is optimal (up to constants) among first-order methods!

Method Selection Guide

- **GD:** Simplest. Use when κ is small or implementation simplicity matters.
- **CG:** Best for quadratics / linear systems. No need to know L, μ .
- **Nesterov AGD:** Best general-purpose accelerated method. Needs L (and μ for SC case).
- **Newton:** Best when n is moderate and Hessian is cheap. Fastest local convergence.

Cost per Iteration

- GD / Heavy Ball / Nesterov: $O(n)$ (one gradient evaluation)
- CG: $O(n)$ for sparse $\mathbf{Q}$, $O(n^2)$ for dense

Summary: What We Proved

1. **Newton:** Local quadratic convergence

$$\|\mathbf{x}_{k+1} - \mathbf{x}^*\| \leq \frac{L_H}{2\mu} \|\mathbf{x}_k - \mathbf{x}^*\|^2$$

2. **CG:** Rate $2 \left(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1} \right)^k$ on quadratics (via Chebyshev polynomials)
3. **Heavy Ball:** Rate $\left(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1} \right)^k$ on quadratics (via spectral radius)
4. **Nesterov AGD:**
 - SC: $(1 - 1/\sqrt{\kappa})^k$ (via Lyapunov function)
 - Convex: $O(1/k^2)$ (via variable momentum)
5. **Lower bounds:** These rates are **optimal** for first-order methods.

Next time: Stochastic gradient descent, proximal methods, and